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C20-C–105

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BOARD DIPLOMA EXAMINATION, (C-20)

MARCH/APRIL—2025

DCE – FIRST YEAR EXAMINATION

ENGINEERING MECHANICS

Time : 3 Hours] [*Total Marks* : 80

PART—A

3×10=30

Instructions : (1) Answer **all** questions.

(2) Each question carries **three** marks.

(3) Answers should be brief and straight to the point and shall not exceed five simple sentences.

1. Define the SI unit and symbol for the following :

(a) Stress

(b) Power

(c) Force

2. State the law of parallelogram of forces.

3. List out the various type of beams.

4. Locate the position of centroid of the following figures with a neat sketch :

(a) Circle

(b) Triangle

(c) Rectangle

5. Define polar moment of inertia of a plane area.

6. Define the terms *(a)* Poisson's Ratio and *(b)* Factory of safety.

7. The Bulk modulus of a material is 125 GPa and Young's Modulus is 200 GPa. What is its Poisson's Ratio.

8. Draw the stress-strain diagram for a mild steel specimen subjected to a tensile force and indicate all the salient points.

9. Define the terms *(a)* Shear force and *(b)* Bending moment.

10. A simply supported beam of 5 m long has a point load of 30 kN at its centre. Determine the maximum shear force and bending moment.

PART—B

8×5=40

Instructions : (1) Answer either *(a)* **or** *(b)* from each questions.

(2) Each question carries **eight** marks.

(3) Answers should be comprehensive and criterion for valuation is the content but not the length of the answer.

11. *(a)* Calculate the forces in the ropes *AB* and *AC* of the arrangement as shown in Figure A.

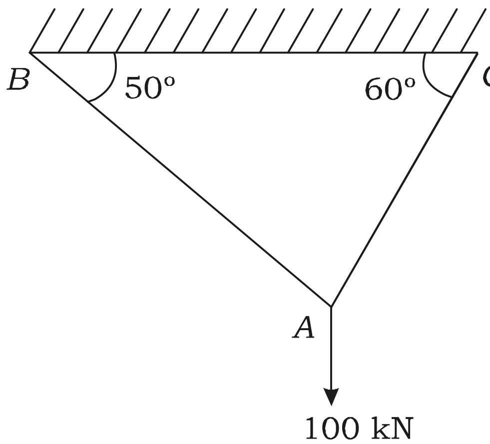

Figure A

(OR)

(b) Find the magnitude and direction of resultant force of the following coplaner concurrent.

(i) 100 kN due North

(ii) 50 kN at 30° in the direction of North of East

(iii) 60 kN at 45° in the direction of South of West

(iv) 45 kN at 60° in the direction of North of West

(v) 80 kN at 25° in the direction of South of East

12. *(a)* Locate the centroid of an equal angular section of size 120 mm × 120 mm × 10 mm.

(OR)

(b) Determine the centroid of I-section having the following dimensions :

Bottom flange = 300 mm × 100 mm

Top flange = 150 mm × 50 mm and Web = 400 mm × 50 mm

13. *(a)* Find the moment of inertia about of a T-section having flange 150 mm × 50 mm and web 50 mm × 150 mm about X-X axis and Y-Y axis through CG of the section.

(OR)

(b) Find the radius of gyration of a hollow circular section of external diameter 300 mm and internal diameter 200 mm.

14. *(a)* A bar of steel of diameter 16 mm and length 600 mm is subjected to axial force as shown in Figure B. If E for steel is 210 kN/mm^2. Determine the change in its length.

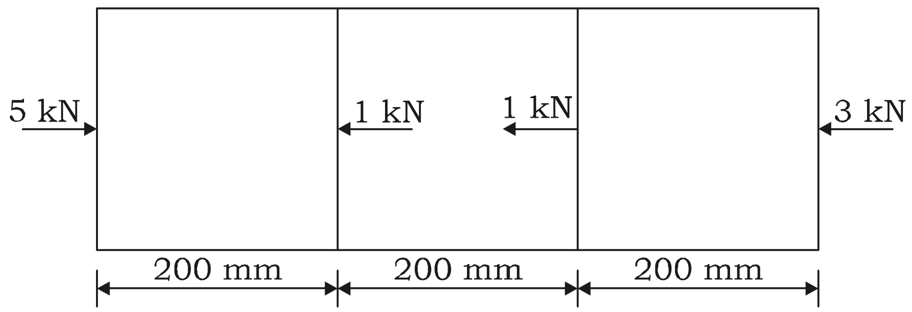

Figure B

(OR)

(b) A load of 500 N is applied on a steel wire of dia 2 mm and length 1 m. Calculate the strain energy stored in the material, if the same load is suddenly applied. Calculate the strain energy absorbed. Take $E = 2 \times 10^5$ N/mm^2.

15. *(a)* A horizontal beam of 12 m long simply supported at its ends, is subjected to vertical loads of 10 kN, 20 kN and 25 kN at 3 m, 7 m and 10 m from left support respectively. Draw shear force and bending moment diagrams indicating values at salient points.

(OR)

(b) A 5 m cantilever carries a point load of 20 kN at the free end and the u.d.l. of 10 kN/m over a length of 3 m from the fixed end. Draw shear force and bending moment diagrams.

PART—C

10×1=10

Instructions : (1) Answer the following question.

(2) The question carries **ten** marks.

(3) Answer should be comprehensive and the criterion for valuation is the content but not the length of the answer.

16. An over hanging beam of 9 m length is simply supported at 6 m from left end. It carries a u.d.l. of 5 kN/m over the supported length and concentrated load of 10 kN at the right extreme end. Sketch the shear force and bending moment diagrams. Locate the point of contra flexure and determine the position and magnitude of maximum bending moment.

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